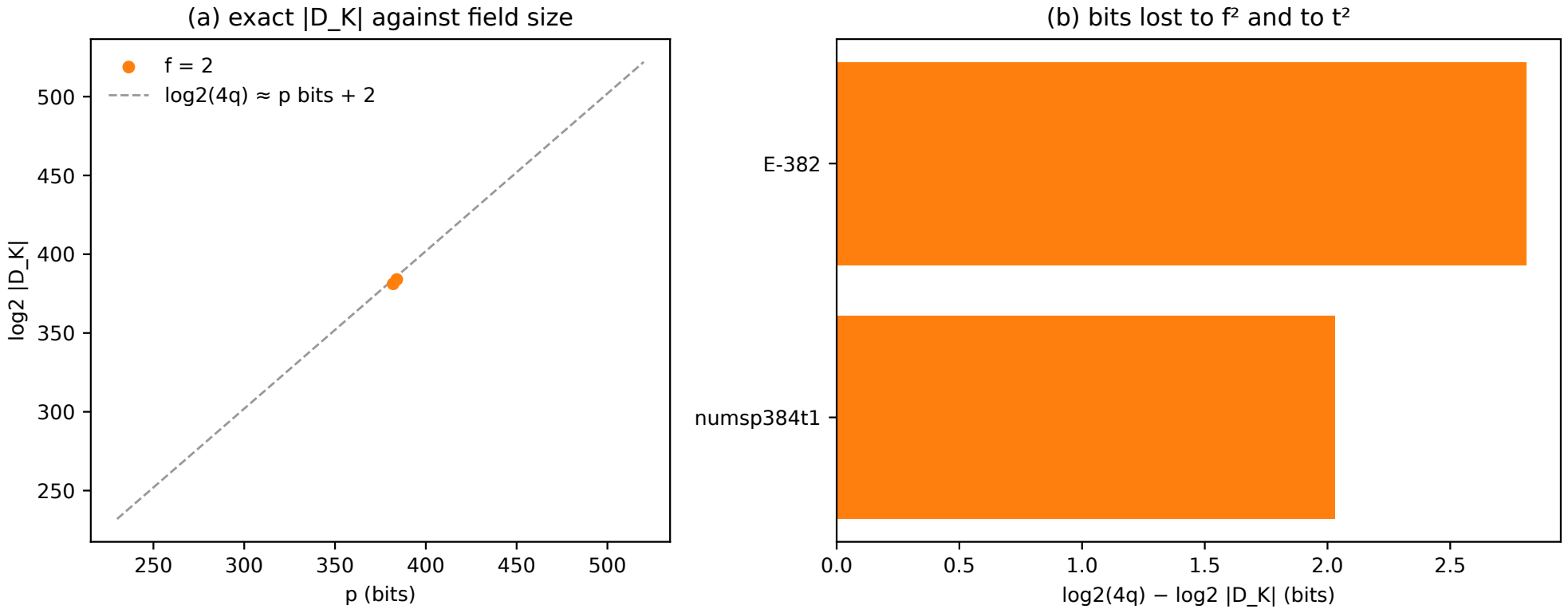
Exact CM discriminants of 3 more standardized curves (batch 3)

research/endosweep_dk_exact_batch3_20261017 (2026-10-17). Data: `dk_exact.json`, `factorizations.jsonl`; std-curves commit `77fe6e3585ca2c2225b59d7df24b7c775437276f`. Every factor of $4q - t^2$ is proved prime by PARI `isprime`. These are exact computations, not measurements; the README states method, scope and regeneration. They are the three curves the factoring driver completed between the second frozen snapshot (batch 2, research/endosweep_dk_exact_batch2_20261016) and the third, each by a CADO-NFS split of a C104, C106 or C109. std-curves records no `cm_disc` for any of the three, so none was checked against the literature. The driver's state file was briefly truncated on disk at about 02:22Z on 2026-10-17 when the disk filled; it was rewritten from the driver's in-memory state before snapshot 3, and every factor here is re-proved prime.



n = 2 curves; exact integers (PARI isprime-proved factorisations), no uncertainty. Not shown: D_K = −3 on arkworks/bw6_761.

Figure 1: (a) $\log_2|D_K|$ against p for the 2 curves with a large D_K, coloured by the conductor f. (b) The bits between $\log_2 4q$ and $\log_2|D_K|$. Recorded values from `dk_exact.json`. Not drawn: $D_K = -3$ on bw6_761.

Results

curve	p bits	4q−t² digits	s digits	f	D_K	log2 |D_K|	min degree	std-curves
other/E-382	382	116	115	2	(115 digits)	381.19	2^379.19	no cm_disc
nums/numsp384t1	384	117	116	2	(116 digits)	383.97	2^381.97	no cm_disc
arkworks/bw6_761	761	230	1	(115 digits)	-3	1.58	1	no cm_disc

Excluded: P-256, P-384, P-521, Curve25519 (research/endosweep_nist_20261006), the 18 curves of research/endosweep_dk_exact_20261016 and the 6 of research/endosweep_dk_exact_batch2_20261016. Lean: `CryptoResearch.Endosweep.DkExactBatch3.exact_discriminants` (target `TASK-20261017-9c045d`) checks, per curve, $q+1-t=hn$, $t^2-4q=D_K f^2$, the product of the factors, the squarefree part and the D_K construction, $D_K \equiv 0,1 \pmod 4$ and the degree bound, reusing `DkExact.DkCert`; fundamentality is conditional on the primality of the factors above 10^8 (PARI, not Lean).